Probabilistic uncertainty

for CFD surrogates

Observation models, posterior prediction, calibration

MIE 690A: AI in Fluid Mechanics

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How to study this lecture

Part A follows supervised regression in Week 2. It requires Gaussian densities, variance and basic differentiation. Part B uses SVD/POD, matrix covariance and least squares; study it after the Week 4 POD bridge. The companion notebook is `notebooks/week02_1/Probabilistic_UQ_CFD.ipynb`. This searchable self-study edition preserves the original fourteen lecture topics and retained experiment, and adds numerical derivations and exercises. The slide deck remains an editable classroom companion. No new CFD or model validation is claimed by this editorial revision.

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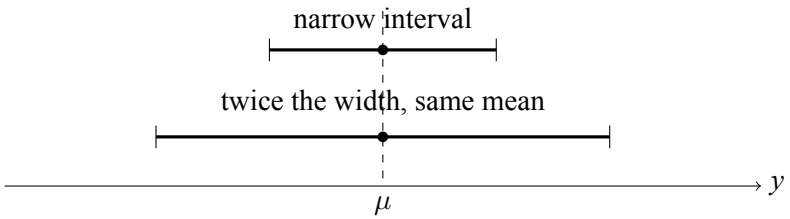
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Part A: observation models and exact Bayesian updating

1 Why one prediction does not express confidence

A deterministic surrogate returns a point prediction $\hat{y}(x)$. Its mean error can be small while the consequences of an individual miss remain serious. For example, an accurate mean wall quantity is insufficient if a design decision depends on whether a local threshold is exceeded. A predictive distribution adds information about plausible values and their probabilities under stated assumptions. It is not a guarantee against an unmodelled physical mechanism.

Two predictions can have the same mean and different spreads. The intervals below share the same axis and mean. The second has twice the width. That geometric statement says nothing yet about whether either interval is calibrated.



Prediction question. Which model would you trust near an operating limit? Width alone cannot answer: inspect accuracy, coverage and the physical mechanisms omitted from the model.

2 What Week 2.1 adds to supervised learning

Week 2 introduced features and targets, scaling, loss functions and complete-case splits. Here the model additionally states an observation process, computes a posterior predictive distribution, scores that distribution and calibrates its width without consulting test cases. No new neural architecture is required. The first worked example is exact linear Gaussian inference. This lets us separate probability assumptions from the optimization difficulties of a large network. The later field example combines a POD representation with Gaussian processes for its coefficients.

By the end, you should be able to explain what is random, compute a scalar posterior by hand, identify the terms in predictive variance, distinguish a marginal interval from a whole-field guarantee, and report a failed coverage target honestly. Running a plotting cell alone does not demonstrate these skills.

3 Three sources of uncertainty

Aleatoric uncertainty describes variability such as measurement noise or unresolved fluctuations. More observations do not necessarily remove the variability of the next observation, though they may estimate its distribution more accurately.

Epistemic uncertainty describes incomplete knowledge of the map or its parameters. Informative new cases can reduce this component within the model's assumptions. Repeated optimizer seeds measure only one restricted source of variation; their spread is not automatically the entire epistemic uncertainty.

Simulator discrepancy describes the difference between the simulator and physical reality, including missing physics and numerical limitations. A surrogate can confidently reproduce a biased CFD solver. A posterior over surrogate coefficients alone does not include this discrepancy.

Do not combine these mechanisms into an unexplained error bar. For every reported interval, write what random experiment or uncertainty source it covers. In particular, noise added to a retained CFD field is an artificial observation experiment, not an independent new CFD realization.

4 A Gaussian likelihood and its complete loss

Suppose independent observations satisfy

$$y_i = f_\theta(x_i) + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \sigma^2).$$

The likelihood is a product of Gaussian densities. Taking its negative logarithm gives

$$-\log p(\mathbf{y} \mid \theta, \sigma^2) = \frac{n}{2}\log(2\pi) + n\log\sigma + \frac{1}{2\sigma^2}\sum_{i=1}^{n}[y_i - f_\theta(x_i)]^2.$$

If σ is known and fixed, minimizing this expression with respect to θ is equivalent to minimizing MSE. This is one probabilistic interpretation of squared-error fitting, not a proof that all physical errors are Gaussian. If σ is learned, $n\log\sigma$ is essential. Omitting it would reward making the variance arbitrarily large. With varying local noise, a heteroscedastic model needs a separate variance and logarithmic penalty for each observation, along with constraints that keep every variance positive.

Numerical likelihood check

For one observation with $y = \mu$, the NLL values for $\sigma = .5, 1, 2$ are approximately $.2258, .9189, 1.6121$. If instead $y - \mu = 1$, they become $2.2258, 1.4189, 1.7371$. The very narrow model is penalized when it confidently misses the observation. The Gaussian density and NLL depend on the units of y; compare scores under the same normalization and state those units.

5 Bayesian linear regression as an exact update

For the linear model $\mathbf{y} = \Phi\mathbf{w} + \boldsymbol{\epsilon}$, let $\mathbf{w} \sim \mathcal{N}(\mathbf{m}_0, S_0)$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2 I)$, independently. If there are n observations and d coefficients, Φ is $n \times d$, $\mathbf{w}$ is $d \times 1$ and S_0 is $d \times d$. Bayes' rule multiplies prior and likelihood. Expanding their quadratic forms and collecting terms in $\mathbf{w}$ gives

$$\begin{aligned} S_n^{-1} &= S_0^{-1} + \sigma^{-2}\Phi^T\Phi, \\ \mathbf{m}_n &= S_n(S_0^{-1}\mathbf{m}_0 + \sigma^{-2}\Phi^T\mathbf{y}), \\ \mathbf{w} \mid \mathbf{y} &\sim \mathcal{N}(\mathbf{m}_n, S_n). \end{aligned}$$

Posterior precision equals prior precision plus data precision. This expression also identifies why directions poorly excited by Φ remain uncertain. Use linear solves or a factorization in code rather than explicitly inverting large matrices. All covariance matrices must be symmetric and physically and numerically consistent with their declared units.

Poiseuille amplitude with two observations

Write a normalized parabolic velocity field as $u(y) = w\phi(y)$, where the known shape ϕ is fixed and the unknown scalar amplitude has prior $w \sim \mathcal{N}(1, .25^2)$. For a simple hand calculation, use two normalized measurements with unit basis values: $\Phi = (1, 1)^T$, observations $(.9, 1.1)^T$ and known $\sigma = .1$. Then

$$S_n^{-1} = 16 + 200 = 216, \qquad m_n = (16 + 90 + 110)/216 = 1, \quad S_n = 1/216.$$

The notebook uses the same principle with a larger sensor set; its result is not a nonlinear turbulence-model inference. As an exercise, discard the second measurement. The posterior precision is 116,

mean $106/116 = .913793$ and variance $1/116$. Less data increase parameter uncertainty but do not change the declared variance of the next measurement's noise.

6 Predictive uncertainty has two terms

For a new basis vector ϕ_*, the latent prediction $f_* = \phi_*^T \mathbf{w}$ has mean $\phi_*^T \mathbf{m}_n$ and variance $\phi_*^T S_n \phi_*$. A noisy future observation adds a separate noise term:

$$\mathrm{Var}(y_* \mid x_*, \mathcal{D}) = \underbrace{\sigma^2}_{\text{observation noise}} + \underbrace{\phi_*^T S_n \phi_*}_{\text{parameter uncertainty}} .$$

For the two-measurement example with $\phi_* = 1$, latent variance is $.00462963$ and predictive observation variance is $.01462963$. The Gaussian 95% predictive interval is approximately $1 \pm .2371$. The latent interval is narrower. Neither interval includes simulator discrepancy unless a separate model for it has been added. If $\phi_* = 0$ at a prescribed boundary, coefficient uncertainty contributes zero there; that does not prove all physical boundary uncertainty is zero.

Checkpoint. State whether a plotted band is for latent velocity, noisy observation, a pointwise quantity, or an entire field. Explain what changes when observations are correlated. Replacing a diagonal noise covariance with a full matrix changes the precision update; counting correlated samples as independent can produce unjustified confidence.

Part B: field uncertainty after the POD prerequisite

7 POD compresses field uncertainty

Arrange complete training fields into snapshot rows, subtract the training mean and compute an SVD. Retain r spatial modes in Φ_r:

$$\mathbf{u}(Re) \approx \overline{\mathbf{u}} + \Phi_r \mathbf{a}(Re).$$

The input Reynolds number is a case parameter, and the coefficients are functions of that parameter. A GP for each retained coefficient predicts a distribution rather than a single coefficient. A fixed low-rank basis reduces a large field problem to a small number of probabilistic regression problems. It does not make omitted modes disappear from the physical error.

For training coefficient values $\mathbf{a}$ with zero prior mean, covariance matrix K and noise variance τ^2, a test cross-covariance vector k_* and prior variance k_{**} give

$$\mu_* = k_*^T (K + \tau^2 I)^{-1} \mathbf{a},$$
$$s_*^2 = k_{**} - k_*^T (K + \tau^2 I)^{-1} k_*.$$

The second expression is latent variance. Add τ^2 when predicting a noisy coefficient observation under that model. Kernel amplitude has coefficient-squared units; length scale has input units. Scaling the input changes the numerical meaning of a length scale, so report both normalization and kernel settings.

A two-observation GP, computed by hand

Let $K = \left[\begin{smallmatrix} 1 & .5 \\ .5 & 1 \end{smallmatrix}\right]$, $\tau^2 = .25$, observations $(1, -1)^T$, $k_* = (.5, .5)^T$ and $k_{**} = 1$. Solving $(K + .25I)\alpha = \mathbf{a}$ gives $\alpha = (4/3, -4/3)^T$, hence $\mu_* = 0$. Solving the same matrix against k_* gives $(2/7, 2/7)^T$, so $s_*^2 = 1 - 2/7 = 5/7$. Observation variance would be $5/7 + 1/4 = 27/28$. Explain why opposite observed coefficients give a zero mean at a symmetric query while leaving nonzero uncertainty.

Field covariance propagates as $\mathrm{Cov}(\mathbf{u}_*) = \Phi_r \,\mathrm{Cov}(\mathbf{a}_*) \Phi_r^T$. Independent coefficient GPs omit cross-coefficient posterior dependence; retaining only a field covariance diagonal also omits spatial correlation. Pointwise marginal intervals therefore do not constitute a simultaneous whole-field coverage guarantee. Truncation error, CFD discretization error and model discrepancy remain additional limitations.

8 Calibration stops before the test gate

The retained author protocol used development cases $Re = 100, 150, 200, 225, 250, 350, 400$, a calibration case $Re = 300$, and held-out cases $Re = 175, 275, 375$. Fit the mean/basis/kernel only with their permitted development data. Use the calibration case to select an interval scale, then freeze it. The statistical split unit is a complete Reynolds case, not random grid nodes from the same field.

A constant multiplier on standard deviation can repair a scale mismatch on validation. It cannot repair every missing physical mechanism, a wrongly shaped uncertainty field or distribution shift. A single calibration case also gives limited evidence about transfer across parameter space. Record the candidates, criterion, selected scale and time of freezing before opening the test table.

Public reuse and truly untouched assessment

The retained test fields are now visible in the course and in earlier lessons. Their exclusion from fitting remains meaningful, but they are public held-out regression cases for a continuing student. A new blind student claim requires new instructor-held fields revealed only after a frozen protocol. Otherwise report a reproducible public-case study. Later diagnostic changes motivated by observing test failures are post-hoc analyses, not repairs validated on that same untouched test. The notebook labels its later diagnostic extension accordingly.

9 Accuracy, calibration and sharpness

Accuracy asks whether predictive mass lies near truth. Calibration asks whether nominal levels match empirical frequencies in the declared evaluation setting. Sharpness asks whether intervals are as narrow as justified. Optimizing only one can produce an unhelpful distribution. A very wide interval may cover nearly all observations but provide little discrimination; a narrow interval may miss confidently. Report all three rather than a single attractive band.

For a nominal central Gaussian level $1-\alpha$, the marginal interval is $[\mu - z_{1-\alpha/2}s, \mu + z_{1-\alpha/2}s]$. Empirical pointwise coverage is the fraction of evaluated values falling within their corresponding interval. Report nominal level, actual coverage, mean width and case count. Grid nodes share a flow field and are strongly dependent, so their number must not be presented as the number of independent CFD experiments. Casewise summaries and new complete cases strengthen the interpretation.

10 Proper scores for the predictive distribution

For a scalar Gaussian prediction, the negative log likelihood is

$$\mathrm{NLL} = \tfrac{1}{2}\log(2\pi s^2) + \frac{(y-\mu)^2}{2s^2}.$$

A very small s with a wrong mean receives a severe penalty. Ensure the variance is positive and explain any numerical floor; it is not a fitted physical noise model. The continuous ranked probability score compares the predictive CDF F with the observed step CDF:

$$\mathrm{CRPS}(F,y) = \int_{-\infty}^{\infty} \left[F(z) - \mathbf{1}\{y \le z\}\right]^2 dz.$$

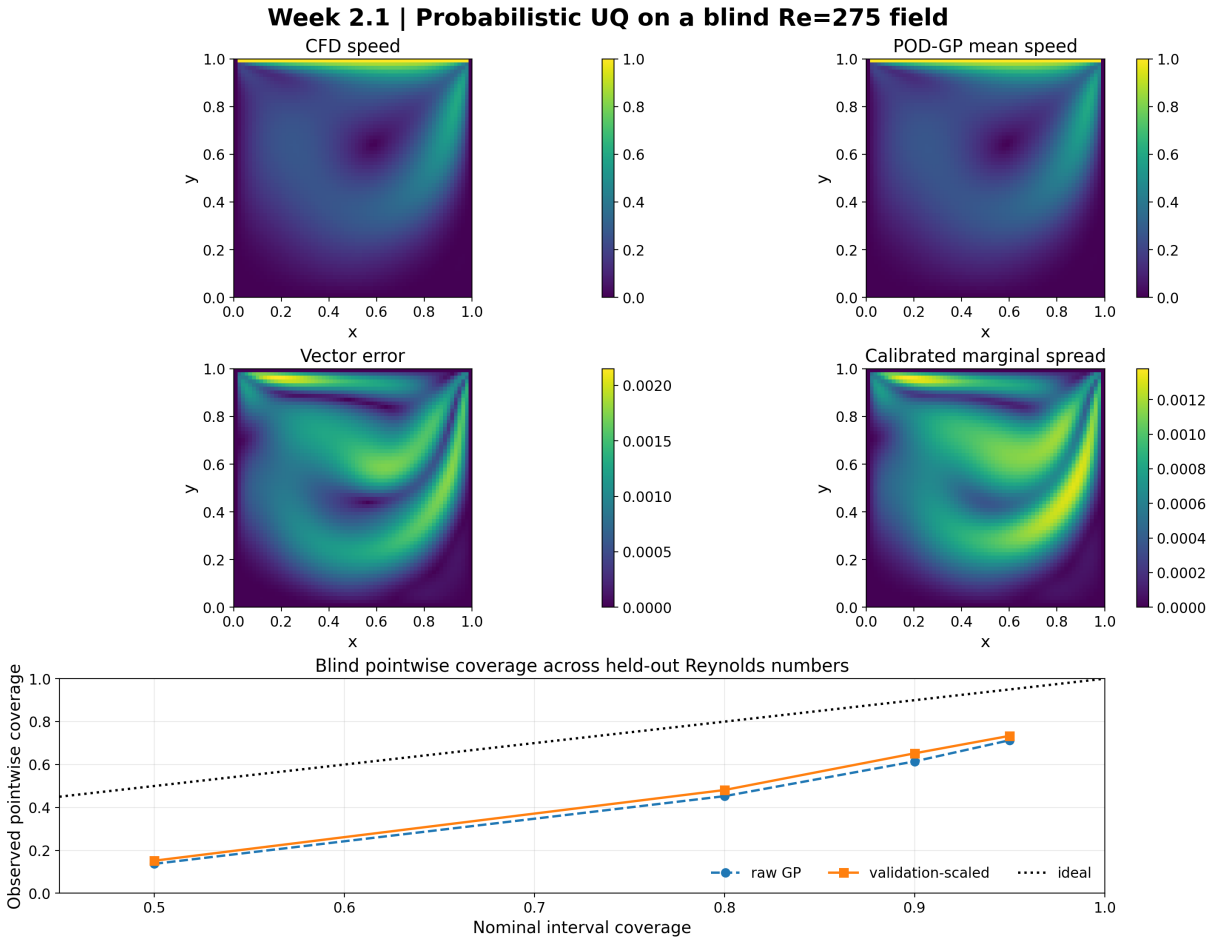
Lower scores are better under the same target definition and units. CRPS has the units of the target. For a Gaussian with $y = \mu$, $\mathrm{CRPS} = s(\sqrt{2}-1)/\sqrt{\pi} \simeq .233695s$. The likelihood example in Part A shows why confident errors matter. Apply both scores to the same cases and pair them with coverage and width. Do not use pointwise scores to silently claim joint distribution calibration.

11 Reading the retained field evidence

The original figure compares the CFD speed field, POD–GP mean speed, vector error and calibrated marginal spread at $Re = 275$, together with coverage curves. The figure on the following full-width evidence page is enlarged without changing its pixels or underlying data. It is retained author evidence, not a rerun. Read the error and spread panels separately: a visually plausible uncertainty pattern is not a demonstrated probability statement.

The retained mean POD–GP relative L_2 error is 0.333%, compared with 0.414% for mean interpolation. At nominal 90%, the calibrated pointwise coverage is 65.2%. The mean improved while coverage remained substantially below the target. Validation-only rescaling increased coverage but did not establish calibration across the held-out Reynolds cases. Preserve this failure in the report; it teaches more than changing the interval until the displayed case looks successful.

Retained CFD uncertainty evidence



Original field panels: held-out $Re = 275$ under the author protocol. Raster scientific evidence is preserved. The searchable equations and exact metric interpretation appear in the surrounding text. Image provenance: `results/probabilistic_uq/`; no solver or fitting run was performed for this re-layout.

12 An uncertainty conclusion with appropriate limits

A defensible conclusion distinguishes mean accuracy from uncertainty reliability: within the retained cavity family, the tested POD–GP mean improves on the specified interpolation comparator, while its nominal 90% marginal intervals remain under-covering. This does not establish model-discrepancy calibration, whole-field coverage or extrapolation safety. Potential future studies include additional independent calibration cases, explicit discrepancy modelling and checks of omitted modal/cross-coefficient covariance. These are proposals, not outcomes demonstrated by the existing figure.

Worked interpretation question

A surrogate has lower relative field error but only 65.2% empirical coverage at nominal 90%. Should it be called a calibrated 90% predictor? No. Report its improved mean and failed coverage target together. If you change the kernel or interval rule after seeing this failure, use fresh untouched cases for a final claim. The old fields may remain a useful diagnostic regression set.

13 Exit checklist, exercises and sources

Before submitting, state what is random and what mechanisms are omitted; list complete physical cases and prior exposure; freeze calibration before test; report accuracy, coverage, width, NLL and CRPS; preserve under-coverage as evidence.

1. Recompute the scalar two-observation posterior and the one-observation variant without the helper. Give latent and noisy predictive variance separately.
2. Recompute the two-observation GP solve, including matrix dimensions and units. Show why a kernel covariance must be positive semidefinite.
3. For predictions with the same mean and $s = .5, 1, 2$, compute NLL at zero and unit residual. Explain why widening every interval cannot optimize all criteria.
4. Explain why 4225 grid nodes from one CFD case are not 4225 independent validation cases, and why pointwise coverage is not simultaneous field coverage.
5. Propose new data that would help distinguish sparse-parameter uncertainty from CFD model discrepancy. Measurements or an independently qualified simulator address a different question from simply adding more points from one field.

The notebook and retained protocol are the primary sources for the numerical experiment. For probability and GP derivations, see Rasmussen and Williams, *Gaussian Processes for Machine Learning*,open textbook. For proper scoring rules see Gneiting and Raftery (2007), proper scoring rules article. For the distinction between parameter uncertainty and model discrepancy, see Kennedy and O'Hagan (2001), model discrepancy article. These references support the concepts; they do not independently validate the course CFD labels or the retained coverage experiment.